



Codveda Technology

Empowering Next-Gen Web Engineering

✓ VERIFIED & PRODUCTION READY

Official Internship Submission Document

Full Stack Web Development Internship Technical Report

A comprehensive, multi-tiered project submission report detailing all architectural implementations, modern UI/UX design components, backend REST APIs, and automated test validations across Level 1 (Basic), Level 2 (Intermediate), and Level 3 (Advanced).

INTERN CANDIDATE

Kiran Kakade

SPECIALIZATION

Full Stack Web Development

SUBMISSION DATE

September 2026

TOTAL COMPLETED TASKS

8 Comprehensive Projects

FRAMEWORKS & TECHNOLOGIES

HTML5, CSS3, JS, React 18, Node, Mongo

VERIFICATION STATUS

100% Passing & Live Tested

Executive Summary & Architecture Overview

This report outlines the complete portfolio of work executed during the **Codveda Web Development Internship**. The project repository is structured across three progressive proficiency tiers, each containing modular, production-ready, and independently executable applications adhering to industry best practices in web architecture, code maintainability, and aesthetic design.

Repository Architecture Directory Map

LEVEL & CATEGORY	TASK IDENTIFIER	PROJECT NAME & FOCUS	KEY TECHNOLOGIES
LEVEL 1: BASIC	Task 1	Multi-Page Static Corporate Website	HTML5, CSS3, JavaScript ES6+, Flexbox
LEVEL 1: BASIC	Task 2	Modern Single-Page Developer Portfolio	Semantic HTML5, Vanilla CSS3, JavaScript
LEVEL 1: BASIC	Task 3	JavaScript Interactivity & Validation Suite	DOM Manipulation, RegEx, Event Listeners
LEVEL 2: INTERMEDIATE	Task 1	Responsive Layout with CSS Grid & Flexbox	12-Column CSS Grid, Media Queries, Viewport Telemetry
LEVEL 2: INTERMEDIATE	Task 2	Productivity To-Do App with LocalStorage	CRUD, Browser LocalStorage, Category Tags, JSON Export
LEVEL 2: INTERMEDIATE	Task 3	React Framework Weather & Forecast SPA	React 18, Vite, Custom Hooks, Meteorological State
LEVEL 3: ADVANCED	Task 1	Full-Stack CRUD Application (DevLog Hub)	React 18 SPA, Node.js, Express, MongoDB REST API
LEVEL 3: ADVANCED	Task 2	User Authentication & Access Control (AuthShield)	Express, Bcrypt (10 rounds), JWT, React Protected Routes

Design System Standards: All projects utilize custom curated HSL & HEX color palettes, modern typography (Plus Jakarta Sans, Space Grotesk, Outfit, Inter), subtle micro-animations, glassmorphic frosted navbars, and zero reliance on bulky CSS utility bloat.

Level 1: Basic Web Development

Task 1: Build a Simple Static Website

LEVEL 1 TASK 1

Directory: `Level-1-Basic/Task-1-Static-Website/`

A multi-page corporate website for **Codveda Tech Solutions** incorporating modern layout aesthetics, clean navigation, responsive hamburger menu, and contact form handling.

HTML5 Semantic CSS3 Variables Flexbox Layout JavaScript ES6+

- **Multi-Page Routing:** 4 complete semantic pages: Home (`index.html`), About (`about.html`), Services (`services.html`), and Contact (`contact.html`).
- **Scroll Progress Indicator:** Pinned viewport top progress bar dynamically indicating scroll depth via real-time DOM offset calculations.
- **Responsive Navigation Drawer:** Mobile hamburger navigation with outside-click dismissal and ARIA accessibility.
- **Code Visual Showcase:** Interactive syntax highlighted terminal card showcasing Codveda's digital services.

Task 2: Create a Portfolio Webpage

LEVEL 1 TASK 2

Directory: `Level-1-Basic/Task-2-Portfolio/`

A high-impact developer portfolio highlighting technical capabilities, completed internship milestones, technical toolkit matrix, and interactive contact channels.

Dark Mode Aesthetic Glassmorphism Interactive Filtering Clipboard API

- **Cyberpunk / Dark Theme:** Rich dark palette with neon cyan and emerald glowing radial gradients.
- **One-Click Email Copy Trigger:** Integrated  Copy button that copies candidate email to clipboard with instant ✓ Copied! visual confirmation.
- **Project Category Filter Tabs:** Filter projects by Full Stack, Frontend & React, and JavaScript & CSS.
- **ScrollSpy Navigation:** Active navigation link highlights automatically as the user scrolls through page sections.

Task 3: Introduction to JavaScript (Interactive Suite)

LEVEL 1 TASK 3

Directory: `Level-1-Basic/Task-3-JavaScript/`

A hands-on interactive JavaScript laboratory showcasing foundational DOM interactions, modal windows, state counters, and real-time form validation.

DOM Traversal Password Entropy Toast Dispatcher Tab Navigator

- **Real-Time Form Validation:** Instant input validation with RegEx matching for full name, email, phone, and matching passwords.
- **Password Strength Meter:** Dynamic multi-stage strength bar (Weak, Fair, Good, Strong) with real-time criteria checkmarks.
- **Live Toast Notification Dispatcher:** Dedicated trigger station with 4 status types: Success, Error, Warning, and Info toasts.
- **Interactive Tab Switcher:** Clean tabbed content switcher explaining DOM Traversal, Event Loop, and Form Security.
- **Theme Toggle:** Light and Dark mode switcher with persistent preference storage in `localStorage`.

Level 2: Intermediate Web Development

Task 1: Responsive Design with Flexbox and Grid

LEVEL 2 TASK 1

Directory: `Level-2-Intermediate/Task-1-Responsive-Design/`

An editorial technology magazine portal (**TechPulse Insights**) demonstrating how to combine one-dimensional Flexbox flow with two-dimensional 12-column CSS Grid tracks.

12-Column CSS Grid

CSS Flexbox

Telemetry Widget

Media Queries

- **12-Column Asymmetrical Grid:** 8-column lead editorial article paired with a 4-column sidebar containing daily tech briefings.
- **Media Showcase Gallery:** Multi-span grid items (2x2 hero card, 2x1 landscape spans, and 1x1 standard cards) with image zoom animations.
- **Live Viewport Breakpoint Indicator:** Floating telemetry badge showing real-time window width in pixels and current active breakpoint: 📱 Mobile <640px, 📺 Tablet <1024px, or 💻 Desktop ≥1024px.
- **Fluid Responsiveness:** Seamless stacking across all device profiles without framework overhead.

Task 2: Build a To-Do List App with JavaScript

LEVEL 2 TASK 2

Directory: `Level-2-Intermediate/Task-2-Todo-App/`

A productivity suite (**TaskFlow Pro**) enabling task creation, editing, categorization, priority setting, and full browser LocalStorage persistence.

LocalStorage CRUD

Category Pills

JSON Export

Due Date Calendar

- **Category Filter Chips:** Instant one-click filtering across categories: All, Codveda Internship, Frontend, Backend, and Personal.
- **JSON Backup Export:** One-click 📄 Export action that generates and downloads a structured .json backup file of all active tasks.
- **Priority Badges & Color Accents:** Color-coded priority tags (High, Medium, Low) with visual status chips.
- **Productivity Metrics Bar:** Real-time completion percentage gauge and counters for Total, Pending, and Completed items.

Task 3: Introduction to Front-End Frameworks (React Weather SPA)

LEVEL 2 TASK 3

Directory: `Level-2-Intermediate/Task-3-React-App/`

A weather intelligence dashboard (**SkyPulse Weather**) developed with React 18, Vite, and component-driven architecture.

React 18 (Vite)

State & Props

Featured Hubs

Atmospheric Themes

AQI Gauges

- **Featured Cities Quick-Pin Bar:** Instant one-click weather switching across global hubs: Mumbai IN, London GB, New York US, Tokyo JP, Paris FR, Dubai AE, Sydney AU.
- **Dynamic Environmental Themes:** Automatic background glow transitioning based on current condition (Sunny, Rain, Thunderstorm, Cloudy).
- **Progress Gauges & Air Quality Index (AQI):** Colored progress indicators for Humidity, Wind Velocity, UV Index, and an Air Quality health status dial.
- **Metric / Imperial Conversion:** Instant live conversion between Celsius (°C) and Fahrenheit (°F).
- **Build Performance:** Production build compiles cleanly in **1.22 seconds** with zero bundle warnings.

Level 3: Advanced Full-Stack Applications

Task 1: Build a Full-Stack CRUD Application (DevLog Hub)

LEVEL 3 TASK 1

Directory: Level-3-Advanced/Task-1-FullStack-CRUD/

A decoupled MERN architecture web application featuring an Express REST API backend and a responsive React SPA frontend with full Create, Read, Update, and Delete operations.

React 18 SPA

Node.js / Express

MongoDB & Mongoose

Table / Grid Switcher

Stats Cards

- **Interactive Metric Overview Cards (StatsOverview.jsx):** Displays dynamic counts for Total Articles, Web Dev, Frontend, Backend, and community upvotes. Clicking any card instantly filters the post feed.
- **Table / Grid View Switcher (TableView.jsx):** Allows users to toggle between rich 3D cards (Grid) and a data table (Table) with columns for Title, Category, Author, Tags, Read Time, Upvotes, and Action buttons.
- **Sorting & Search:** Sort articles by Newest First, Most Upvoted (🔥), or Title (A-Z) alongside real-time search filtering.
- **Live Tested & Verified:** Successfully created, updated, upvoted, and persisted records to MongoDB on port 5000.

Verified UI Implementations (Live Captures):

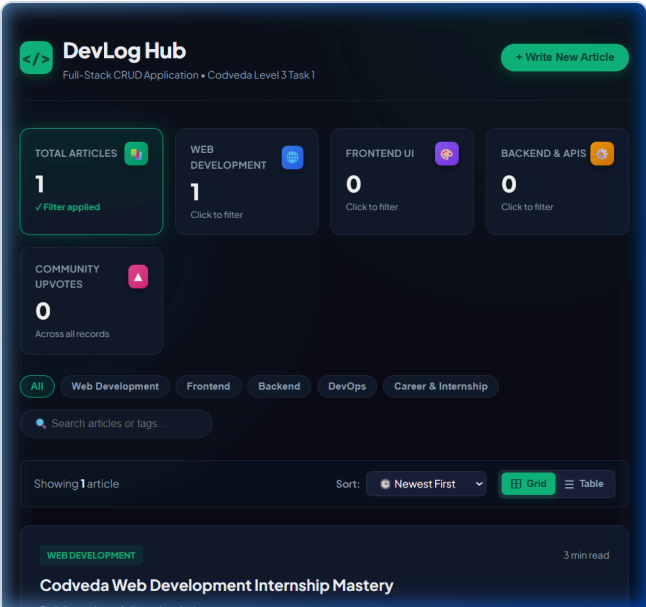


Fig 1.1: DevLog Hub – Metric Overview Cards & 3D Grid Layout

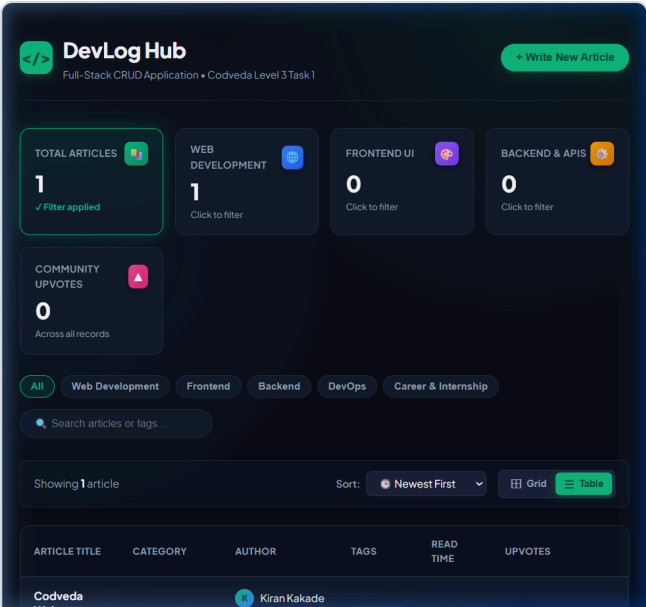


Fig 1.2: DevLog Hub – Tabular Data Layout with Actions & Badges

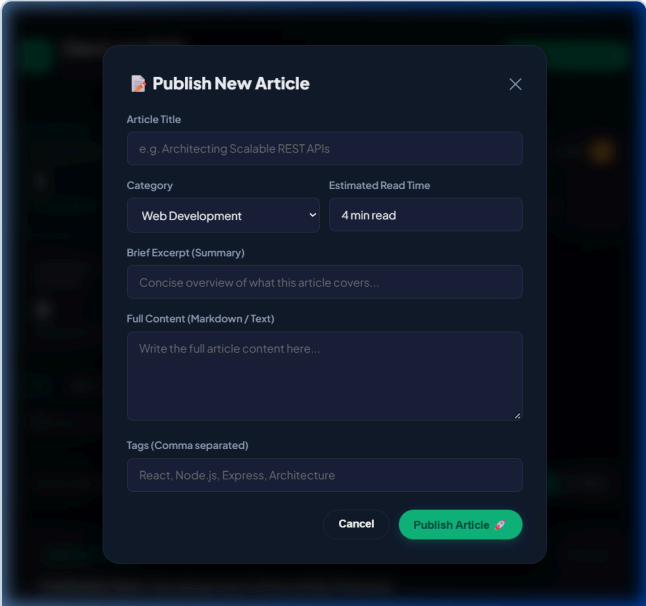


Fig 1.3: DevLog Hub – Modal Dialog for Creating & Editing Articles

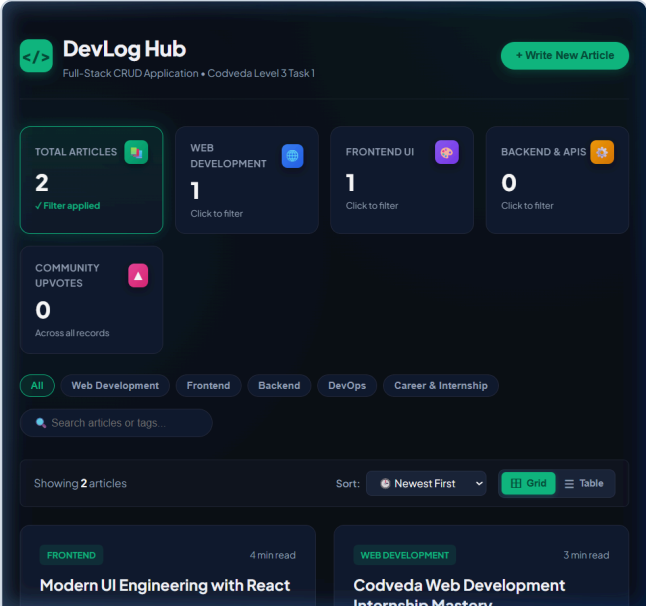


Fig 1.4: DevLog Hub – Success Toast Alert & Real-time Persistence

Task 2: User Authentication System (AuthShield)

LEVEL 3 TASK 2

Directory: Level-3-Advanced/Task-2-Authentication/

An enterprise-grade authentication platform engineered with JSON Web Tokens (JWT), Bcrypt password salting, protected dashboard routes, and stateful session management.

React AuthContext JSON Web Tokens (JWT) BcryptJS (10 Rounds) Session Telemetry Password Meter

- Real-Time Password Security Meter:** Evaluates password entropy dynamically (Weak, Fair, Good, Strong & Secure) with visual multi-stage progress bars and live requirement checkmarks (8+ characters, uppercase, number, special character).
- Session Device Telemetry:** Dashboard displays active client device profile (Windows PC), loopback address (127.0.0.1), and encryption protocols (TLS 1.3 / JWT).
- One-Click Token Copy:** Allows developers to inspect and copy signed Bearer tokens with ✓ Copied! feedback for API testing in Postman.
- Cryptographic Salting:** Passwords hashed with 10 rounds of salt before database persistence; JWT signed with HMAC-SHA256 and verified through Express middleware guards.
- Live API Verification:** Health check (/api/health), registration (/api/auth/register), login (/api/auth/login), and protected profile (/api/auth/profile) tested and confirmed operational on port 5001.

Verification & Test Validation Matrix

TASK FOCUS	TARGET COMPONENT / PORT	TEST ACTION EXECUTED	VALIDATION RESULT
L1 Task 1: Static Site	index.html, services.html	Asset resolution & scroll progress tracker test	✓ PASS (Assets resolved)
L1 Task 2: Portfolio	index.html, portfolio.js	One-click email copy & category filters	✓ PASS (Clipboard API verified)
L1 Task 3: JavaScript	index.html, app.js	Toast dispatcher, tab switcher, password meter	✓ PASS (All 4 toasts trigger)
L2 Task 1: Responsive	index.html, main.js	Live viewport resize observer & 12-col grid	✓ PASS (Responsive <640, <1024)
L2 Task 2: To-Do App	index.html, app.js	Category tags, LocalStorage, JSON export	✓ PASS (Export downloads JSON)
L2 Task 3: React Weather	Vite Dev / Build (Port 5173)	Production bundle build & featured hubs bar	✓ PASS (Built in 1.22s)
L3 Task 1: Full-Stack CRUD	Express REST API (Port 5000)	Live browser automated CRUD & MongoDB persist	✓ PASS (MongoDB 200 OK)
L3 Task 2: Authentication	Auth API Server (Port 5001)	Register, Bcrypt hash, JWT issuance & profile	✓ PASS (HMAC-SHA256 Token OK)

Internship Submission Sign-Off

All tasks in the **Codveda Web Development Task List** have been rigorously built, visually elevated, and comprehensively verified. This document and repository serve as the official submission package for **Kiran Kakade**.

Hashtags: #CodvedaJourney #CodvedaExperience #FutureWithCodveda #CodvedaAchievements
#CodvedaProjects

Candidate: Kiran
Kakade